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Unseen seeds, green minds in the digital era: How SMEs foster innovation through green intellectual capital, AI, and resource commitment

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ABSTRACT

The rapid acceleration of digital transformation has profoundly reshaped societal operations, with Artificial Intelligence (AI) and the Metaverse emerging as key drivers of change. The strategic integration of tacit knowledge and advanced technologies is crucial for enhancing innovation performance and maintaining a competitive advantage in the global economy. This study explores how tacit knowledge sharing influences innovation performance (IP), with a focus on green intellectual capital (GIC), AI, and resource commitment (RC). To examine these relationships, a survey was conducted in three phases among small and medium-sized manufacturing firms in Canada and the UAE, collecting data from 317 UAE and 383 Canadian SMEs. The findings show that GIC plays a key role in linking tacit knowledge sharing to IP, while AI influences how tacit knowledge interacts with GIC, which consists of green human, relational, and structural capital. Additionally, RC strengthens the connection be-tween GIC and IP, reinforcing the importance of resource investment in sustainability and innovation. The originality of this study comes from exploring how AI influences the connection between tacit knowledge and GIC, as well as how RC impacts the link between GIC and IP. The study also recognizes its limitations and outlines areas for future research, encouraging further research into the role of knowledge and technology in driving innovation.

Notation list

AI	Artificial intelligence
GIC	Green intellectual capital
TK	Tacit knowledge sharing
IP	Innovation performance
RC	Resource commitment
KBV	Knowledge based view
TA	Tacit Knowledge awareness
GSC	Green Structural Capital
GHC	Green Human capital
GRC	Green relational capital

1. Introduction

The digital transformation has fundamentally reshaped daily life, influencing how individuals live, work, and how policies are formulated and implemented (Al-Emran, 2023; Melović et al., 2020). This transformation has given rise to two prominent technologies: Artificial Intelligence (AI) and the Metaverse (Jafar et al., 2023; Kautish et al., 2025). To effectively adapt to these changes, it is crucial to embrace these technologies rather than resist them, as they form the foundation of new knowledge-based strategies for the evolving "new normal" (Alinasab et al., 2022; Grybauskas et al., 2022; La Sala et al., 2024). Firms increasingly recognize tacit knowledge as a vital resource for

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